EXPLAINABLE AI POWERED ORGAN TRANSPLANT COORDINATION

A Project Report

U23ADS705 – MINI PROJECT

Submitted by

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Declaration by the Student(s)

We hereby declare that the project titled **EXPLAINABLE AI POWERED ORGAN TRANSPLANT COORDINATION** is an original work carried out by us under the guidance of **Mr. A. S. Syed Fiaz.** This project has not been submitted elsewhere for the award of any degree.

We affirm that all sources have been properly acknowledged and cited. This project supports SDG 3 (Good Health and Well-Being) and SDG 9 (Industry, Innovation and Infrastructure) through AI-based organ matching and explainable recipient prioritization. It is an academic prototype and is not intended for actual clinical or organ allocation decisions.

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ABSTRACT

The need for timely organ transplantation in patients with end-stage organ failure is a critical issue and organ allocation is difficult because of the lack of enough donor organs. The system we built is called Explainable AI Powered Organ Transplant Coordination and it uses a combination of rule-based pre-processing for eligibility, XGBoost for the recipient ranking, SHAP for explainability and human oversight. This system stores information on the donors and recipients, determines the eligibility of recipients based on the conditions defined in the system, and calculates the priority scores according to the conditions of clinical urgency, waiting time, distance, hospital readiness. It also enables email notifications, offers to accept or reject offers and tracks your clinical status. The prototype was created with the help of React.js, TypeScript, FastAPI, SQLAlchemy and SQLite and tested on synthetic data with 356 match runs, with 56 candidates eligible. The XGBoost model achieved an MAE of 0.0202, RMSE of 0.0310, and R^2 of 0.9231. The system is designed for academic and research. It is not used as a substitute for clinical decision making or official organ allocation systems. Enhancement features to be implemented in the future are hospital-based authentication, assessment of organs from deceased donors, tracking of expiration time, and location-based coordination.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ML	Machine Learning
XGBoost	Extreme Gradient Boosting
SHAP	SHapley Additive exPlanations
API	Application Programming Interface
UI	User Interface
UX	User Experience
DB	Database
SQL	Structured Query Language
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
REST	Representational State Transfer
JSON	JavaScript Object Notation
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
TS	TypeScript
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
MLP	Machine Learning Pipeline

1. INTRODUCTION

1.1 Background of the Project

Patients with end-stage organ failure can be treated with organ transplantation, which is a life-saving treatment. But, as there are fewer donors and more recipients, matching organs is difficult and prioritizing donors is difficult. The suggested ETACS for organ transplants integrates the rule-based eligibility screening, XGBoost-based ranking of recipients, and SHAP explanation for the transparent decision-making process. It also supports donor management, recipient management, organ management, hospital management, email notifications, etc. and offers the option to accept or reject offers, tracking clinical status, and fallback processing based on ranking. This is an academic prototype that is accessible via the web, but does not supplant clinical decisions by qualified healthcare professionals.

1.2 Problem Statement

Current organ matching protocols are based on scoring systems and manual review, and can be labor-intensive if multiple candidates are eligible for organs. However, while machine learning models can identify candidates for screening, they may not be interpretable and rule-based screening can only identify eligible candidates, it cannot prioritize them. The proposed system is structured and transparent to support clinical decision and hospital coordination, where the existing rule-based eligibility screening, XGBoost-based ranking, SHAP explanations, and hospital notifications complement each other.

1.3 Motivation

The shortage of donor organs is a major challenge in transplantation, making the systematic identification and prioritization of potential recipients essential. A transparent and organized process is needed to evaluate candidates using relevant clinical and administrative information. Manually evaluating multiple candidate records, maintaining consistent assessments, and ensuring timely hospital follow-up can be challenging. Similarly, users may find machine learning predictions difficult to understand when the factors influencing the results are not clearly explained.

Artificial intelligence and explainable machine learning have the potential to support recipient prioritization. This project aims to understand how these technologies can improve the evaluation process without replacing medical experts. The proposed system demonstrates an organized and traceable academic prototype that combines rule-based filtering, XGBoost-based ranking, SHAP explanations, and hospital workflow management.

1.4 Objectives

The main objective of this project is to develop a web-based application that demonstrates organ matching, recipient eligibility screening, and AI-based recipient prioritization.

The specific objectives are:

- To develop a web-based platform for managing donor, organ, recipient, and hospital information.
- To apply predefined clinical and administrative eligibility criteria to identify potentially eligible recipients.
- To generate priority scores and rank eligible recipients using the

XGBoost model.

- To integrate SHAP for explaining the factors influencing model predictions.
- To automate email notifications to hospitals.
- To support offer acceptance and rejection and enable manual review of recipient rankings.
- To implement a rank-based fallback process when a higher-ranked candidate cannot proceed.
- To maintain consistent records of activities and decision-related information in a relational database.

The proposed system is an academic decision-support prototype. It is not a substitute for clinical judgment, approved organ allocation guidelines, or decisions made by authorized medical professionals.

1.5 Scope of the Project

The project covers donor management, recipient management, organ management, rule-based eligibility screening, XGBoost-based recipient ranking, SHAP explanations, hospital email notifications, and human review. It considers selected clinical and administrative criteria to generate priority scores for eligible candidates. The system is developed using React.js, TypeScript, FastAPI, SQLAlchemy, and SQLite and is evaluated using synthetic data.

This project is strictly an academic prototype and is not intended for actual organ allocation. Future enhancements include hospital-based authentication, deceased donor assessment, organ preservation-time monitoring, and location-based coordination.

2. LITERATURE REVIEW

2.1 Introduction

In this chapter, studies related to organ matching, organ allocation, transplant outcome prediction, machine learning, explainable artificial intelligence, and ethical decision-making are reviewed. These studies discuss different approaches used to improve donor-recipient compatibility, predict transplant outcomes, prioritize recipients, and support fair organ allocation. The reviewed literature includes traditional allocation methods, machine-learning-based prediction models, deep-learning approaches, and explainable AI techniques. These studies provide the research background for the proposed Explainable AI Powered Organ Transplant Coordination System.

The findings discussed in this chapter are based on the respective published studies and are not the experimental results of the proposed project. The studies are presented in alphabetical order according to the first author's surname, followed by the corresponding numbered references.

2.2 Review of Related Studies

Allgaier et al. [1], in "How does the model make predictions? A systematic literature review on the explainability power of machine learning in healthcare", reviewed different approaches used to explain machine-learning predictions in healthcare. Their study highlights the importance of making model decisions understandable, particularly in applications where predictions may influence important medical decisions.

Ali et al. [2], in "Artificial intelligence assisted risk prediction in organ transplantation: a UK Live-Donor Kidney Transplant Outcome Prediction tool", presented an AI-assisted approach for predicting outcomes following live-donor kidney transplantation. Their research

demonstrates how machine learning can support the assessment of transplant-related risks and provide additional information for clinical decision-making.

Alowidi et al. [3], in "Advancing Kidney Transplantation: A Machine Learning Approach to Enhance Donor-Recipient Matching", investigated the use of machine learning to improve donor-recipient matching in kidney transplantation. Their work explores how patient and donor characteristics can be used to identify suitable transplant matches.

"Artificial intelligence–driven kidney organ allocation: systematic review of clinical outcome prediction, ethical frameworks, and decision-making algorithms" [4], published in BMC Nephrology, reviewed AI-based methods used in kidney organ allocation. The study discusses clinical outcome prediction, ethical considerations, and algorithmic approaches for supporting organ allocation decisions.

Börner et al. [5], in "A Novel Deep Learning Model as a Donor-Recipient Matching Tool to Predict Survival after Liver Transplantation", proposed a deep-learning model for donor-recipient matching in liver transplantation. Their research examined the use of donor and recipient information to predict post-transplant survival and support the identification of suitable transplant combinations.

Bunnik [6], in "Ethics of allocation of donor organs", discussed the ethical challenges involved in distributing donated organs among patients. The study highlights the importance of fairness, medical urgency, transparency, and ethical principles when determining organ allocation priorities.

Dale et al. [7], in "Inconsistent values and algorithmic fairness: a review of organ allocation priority systems in the United States", reviewed organ allocation priority systems and examined differences in

the values used to determine recipient priority. Their work discusses how inconsistent allocation principles can affect fairness and emphasizes the need for transparent and consistent decision-making.

Deshpande [8], in "Smart match: revolutionizing organ allocation through artificial intelligence", discussed the application of artificial intelligence in organ allocation. The study explores how AI-based approaches can support donor-recipient matching and improve the organization of allocation decisions.

Ding et al. [9], in "FairAlloc: Learning Fair Organ Allocation Policy for Liver Transplant", introduced a learning-based approach for developing fair organ allocation policies in liver transplantation. Their work focuses on incorporating fairness into allocation decisions while considering the requirements of transplant prioritization.

"Ensemble-Based Predictive Modeling Machine Learning Framework for Live-kidney Donation Prediction" [10], published in Procedia Computer Science, presented an ensemble-based machine-learning approach for live-kidney donation prediction. The study highlights the use of multiple predictive models to analyze donation-related information and support prediction tasks.

Gnanasambandan and Balasubramanian [11], in "A data-driven framework for fair and efficient organ transplantation using gradient boosting and adaptive genetic allocation", proposed a data-driven framework combining gradient boosting and adaptive genetic allocation. Their research focuses on improving the efficiency and fairness of organ transplantation through machine-learning-based prediction and allocation strategies.

Lebret [12], in "Allocating organs through algorithms and equitable access to transplantation—a European human rights law approach", examined organ allocation algorithms from a European human rights

perspective. The study discusses equitable access to transplantation and emphasizes that algorithmic allocation should follow ethical and legal principles.

Li et al. [13], in "FERI: A Multitask-based Fairness Achieving Algorithm with Applications to Fair Organ Transplantation", proposed a multitask-based algorithm to address fairness in organ transplantation. Their research investigates how fairness-aware machine-learning methods can be applied to allocation problems.

"Machine Learning Algorithms in Controlled Donation After Circulatory Death Under Normothermic Regional Perfusion: A Graft Survival Prediction Model" [14], published in Transplantation, investigated machine-learning algorithms for predicting graft survival in controlled donation after circulatory death under normothermic regional perfusion. The study demonstrates the relevance of predictive modelling in evaluating transplant-related outcomes.

Naqvi et al. [15], in "Donor-Recipient Matching for Kidney Transplantation using Uncertainty Estimation in Generalized Propensity Score", investigated donor-recipient matching using uncertainty estimation and generalized propensity scores. Their approach considers uncertainty in matching decisions, which is important when evaluating possible transplant combinations.

Olawade et al. [16], in "The impact of artificial intelligence and machine learning in organ retrieval and transplantation: A comprehensive review", reviewed applications of AI and machine learning across organ retrieval and transplantation. Their study discusses the potential use of intelligent technologies in supporting transplant-related processes and decision-making.

Pruinelli et al. [17], in "Transforming liver transplant allocation with artificial intelligence and machine learning: a systematic review",

reviewed the application of AI and machine learning in liver transplant allocation. Their research discusses existing predictive approaches, allocation methods, and challenges associated with applying intelligent systems in transplantation.

Rueda et al. [18], in "'Just' accuracy? Procedural fairness demands explainability in AI-based medical resource allocations", examined the relationship between explainability and fairness in AI-based medical resource allocation. Their study emphasizes that prediction accuracy alone is insufficient and that the decision-making process should also be understandable and justifiable.

Salaün et al. [19], in "Predicting graft and patient outcomes following kidney transplantation using interpretable machine learning models", investigated interpretable machine-learning models for predicting graft and patient outcomes after kidney transplantation. Their work demonstrates the importance of combining predictive performance with model interpretability in healthcare applications.

"TransplantPredictor: A deep learning framework for liver transplant matching and survival prediction" [20], published in Results in Engineering, presented a deep-learning framework for liver transplant matching and survival prediction. The study explores the use of predictive models to analyze transplant-related information and support matching and outcome assessment.

2.3 Techniques and Technologies Used in Existing Systems

The reviewed studies demonstrate the use of different techniques in organ transplantation and allocation. Machine-learning algorithms such as gradient boosting, ensemble learning, and other predictive models are used to analyze donor and recipient information and estimate transplant-related outcomes [2], [3], [10], [11], [19].

Deep-learning techniques are also applied to donor-recipient matching

and transplant survival prediction. These approaches use relevant donor and recipient characteristics to identify potential matching combinations and predict possible outcomes [5], [20].

Organ allocation systems also consider medical eligibility, recipient priority, and equitable access to donated organs. Research on algorithmic allocation and fairness discusses the importance of combining medical requirements with transparent and consistent allocation principles [6], [7], [8], [9], [12], [13], [17].

Explainable artificial intelligence is another important area in healthcare prediction. Interpretable machine-learning models and explainability techniques help users understand the factors influencing model predictions. These approaches are particularly relevant when machine-learning outputs are used to support medical resource allocation and transplant-related decisions [1], [18], [19].

The reviewed studies also show that prediction accuracy alone cannot establish the suitability of an organ allocation system. Fairness, uncertainty, clinical considerations, transparency, and human oversight are important when developing AI-assisted transplant coordination systems [4], [6], [14], [15], [16].

2.4 Research Gap

The reviewed studies focus on different aspects of organ transplantation, including donor-recipient matching, transplant outcome prediction, organ allocation, fairness, and explainability. Some studies concentrate on predicting transplant survival, while others investigate allocation policies or the ethical aspects of distributing donated organs.

However, these research areas do not necessarily provide a complete coordination workflow that integrates donor and recipient registration, eligibility verification, recipient ranking, explainable prediction, hospital communication, and offer management within a single

application. The proposed project aims to address this application-level gap by integrating these processes into an Explainable AI Powered Organ Transplant Coordination System. The system uses predefined eligibility criteria to filter potential recipients before applying an XGBoost-based regression model to support recipient priority ranking. SHAP-based explanations are incorporated to provide information about the features influencing the predicted priority scores. The platform also supports hospital-based records, organ and recipient management, notifications, and offer acceptance or rejection. The eligibility filtering and ranking processes are designed to support human review rather than replace clinical judgment or official organ allocation procedures. The proposed system is an academic prototype developed using synthetic data. Its model performance does not establish clinical validity or suitability for real-world organ allocation. Clinical validation, integration with authorized transplant registries, and evaluation using real-world data would be necessary before practical deployment.

2.5 Summary

The literature reviewed in this chapter covers organ matching, transplant outcome prediction, machine learning, deep learning, explainable AI, and fairness in organ allocation. Existing studies demonstrate how intelligent techniques can support donor-recipient matching, predict transplant outcomes, and improve the understanding of allocation decisions. The reviewed research also highlights the importance of transparency, fairness, uncertainty estimation, and human involvement in healthcare decision-making. Based on these studies, the proposed project integrates eligibility filtering, machine-learning-based recipient ranking, SHAP explanations, and hospital coordination features into a single academic decision-support platform.

CHAPTER 1

NEED FOR PROPOSED TECHNIQUE

3.1 Existing Techniques

Current organ-matching procedures typically involve manual recipient selection, rule-based screening, and predefined scoring systems. Although these methods support eligibility assessment, they can be time-consuming when evaluating multiple recipients. They may also have limitations in considering different clinical and administrative factors for prioritization. Machine learning can assist in recipient ranking, while explainability techniques help users understand model predictions. Therefore, an integrated approach involving eligibility screening, recipient ranking, explainability, and human oversight is required.

Table 3.1: Comparison of Existing and Proposed Techniques

Technique	Working Principle	Limitation
Manual Recipient Assessment	Doctors and healthcare professionals evaluate potential recipients individually based on clinical information.	Time-consuming when multiple candidates need to be assessed.
Proposed Integrated Framework	Combines eligibility filtering, XGBoost-based ranking, SHAP explainability, and human review.	Requires further clinical validation before any real-world medical application.

Table 3.1 summarizes the existing methods and the proposed framework. The proposed academic prototype integrates eligibility

screening, recipient prioritization, and explainability while maintaining the importance of human review.

3.2 Limitations of Existing Techniques

Existing organ-matching and recipient-prioritization techniques have several limitations that highlight the need for an integrated approach. Manual assessment can be time-consuming, particularly when multiple potential recipients must be evaluated and their information maintained. Predefined scoring systems may not fully capture the relationships between factors such as clinical urgency, waiting time, and hospital readiness. Furthermore, eligibility screening and recipient prioritization are two different processes. Identifying eligible candidates does not automatically determine their priority order. Machine learning performance also depends on the quality and representativeness of the training data. Incomplete or inadequate data may affect prediction accuracy. Another limitation is the lack of explainability in some machine learning models. Users may find it difficult to understand how predictions are generated and how individual features influence the ranking results. Therefore, automated rankings should not independently determine clinical suitability or replace the judgment of authorized healthcare professionals.

3.3 Proposed Technique

The Explainable AI-Powered Organ Transplant Coordination System is an academic prototype designed to support a structured and transparent organ-matching process. It manages donor, recipient, organ, and hospital information and applies rule-based eligibility screening to identify candidates who satisfy predefined

criteria. Eligible candidates are prioritized using an XGBoost machine learning model based on selected clinical and operational features. The model generates priority scores and ranks the candidates. SHAP explainability is integrated to show how individual features contribute to the predicted scores. The system also supports human review, automated hospital email notifications through SMTP, offer acceptance and rejection, clinical status tracking, and rank-based fallback handling. The technology stack includes React.js and TypeScript for the frontend, FastAPI for the backend, SQLAlchemy for database interaction, and SQLite for data storage. By integrating machine learning, explainable AI, and workflow management, the system aims to improve the organization, transparency, and traceability of the academic organ-matching process. This prototype is developed exclusively for academic and research purposes. It is not intended for actual organ allocation, clinical use, or replacement of decisions made by authorized medical professionals.

3.4 Objectives of the Proposed Technique

The proposed technique aims to organize and improve the transparency and traceability of the organ-matching process through a structured web application. Its main objectives are:

- To maintain donor, recipient, organ, hospital, and matching information in a centralized system.
- To apply predefined eligibility criteria to identify suitable candidates for further assessment.
- To use the XGBoost model to generate priority scores and rank eligible recipients.

- To integrate SHAP explainability to identify the factors influencing the model's priority scores.
- To provide automated email notifications for communication with hospitals.
- To support offer acceptance, rejection, clinical status tracking, and workflow management.
- To implement a rank-based fallback mechanism so that the workflow can continue when a higher-ranked candidate does not proceed.

The overall objective is to demonstrate how explainable machine learning can support recipient prioritization while keeping final decisions under the control of authorized personnel.

3.5 Advantages of the Proposed System

The proposed system combines rule-based eligibility screening, XGBoost-based recipient ranking, and SHAP explainability. It supports the identification of eligible candidates, generates priority scores, and helps users understand the factors influencing model predictions. Automated email notifications and rank-based fallback handling support communication and workflow continuity. Centralized data management provides a structured way to maintain donor, organ, recipient, hospital, and matching information. Human supervision is also incorporated to ensure that automated predictions support, rather than replace, decisions made by authorized medical professionals. Overall, the proposed system improves the organization, traceability, and interpretability of the academic organ transplant coordination process.

CHAPTER 2
DESIGN OF THE PROPOSED SYSTEM AND IMPLEMENTATION

4.1 System Requirements

System requirements define the hardware and software conditions needed for the development, implementation, and testing of the proposed organ-matching and recipient-ranking prototype. These requirements are categorized into two sections: hardware requirements and software requirements.

4.1.1 Hardware Requirements

Table 4.1 summarizes the hardware requirements for developing and executing the application.

Table 4.1: Hardware Requirements

Requirement	Minimum Configuration	Recommended Configuration
RAM	4 GB	8 GB or more
Storage Space	10 GB available	20 GB or more available
Display Resolution	1366 × 768	Full HD (1920 × 1080)
Internet Connectivity	Stable internet connection	Stable broadband connection
Input Devices	Keyboard and mouse	Keyboard and mouse

The hardware requirements for developing and implementing the prototype are presented in Table 4.1. The recommended configuration is adequate for running the machine learning model,

database, backend, and frontend components simultaneously.

4.1.2 Software Requirements

The proposed system uses a combination of frontend, backend, database, and machine learning technologies. Table 4.2 lists the software requirements and their purposes.

Table 4.2: Software Requirements

Software / Technology	Purpose
Windows 10/11, Linux, or macOS	Operating system for development and execution
Python	Backend development and machine learning implementation
JavaScript and TypeScript	Frontend development
React.js	Development of the user interface
FastAPI	Backend framework and REST API implementation
SQLAlchemy	Database operations and Object-Relational Mapping (ORM)
SQLite	Relational data storage
XGBoost	Recipient priority-score prediction
SHAP	Explanation of machine learning predictions
SMTP	Hospital email notifications
Visual Studio Code	Programming and development environment
Git and GitHub	Version control and source-code management
Google Chrome	Application access and testing

Table 4.2 presents the main software technologies used in the implementation. The frontend is developed using React.js and TypeScript, while FastAPI handles backend services and REST API communication. SQLAlchemy and SQLite support structured data management, and XGBoost and SHAP are used for recipient priority prediction and explainability, respectively.

4.2 System Architecture

The proposed system follows a multi-layer architecture consisting of the presentation layer, application layer, machine learning layer, and database layer. The presentation layer is developed using React.js and TypeScript. It provides the user interface for managing donor information, organ details, recipient records, and matching activities. FastAPI is used to implement the application layer. It connects the frontend with the backend, validates input data, applies eligibility rules, processes API requests, and manages workflow operations. The machine learning layer uses XGBoost to calculate priority scores for eligible candidates. SHAP is integrated to explain the contribution of individual features to the model predictions.

The database layer uses SQLite for data storage and SQLAlchemy for database interaction. It maintains organized records of donors, organs, recipients, hospitals, matching activities, offers, and workflow information. An SMTP-based notification component supports email communication with hospitals.

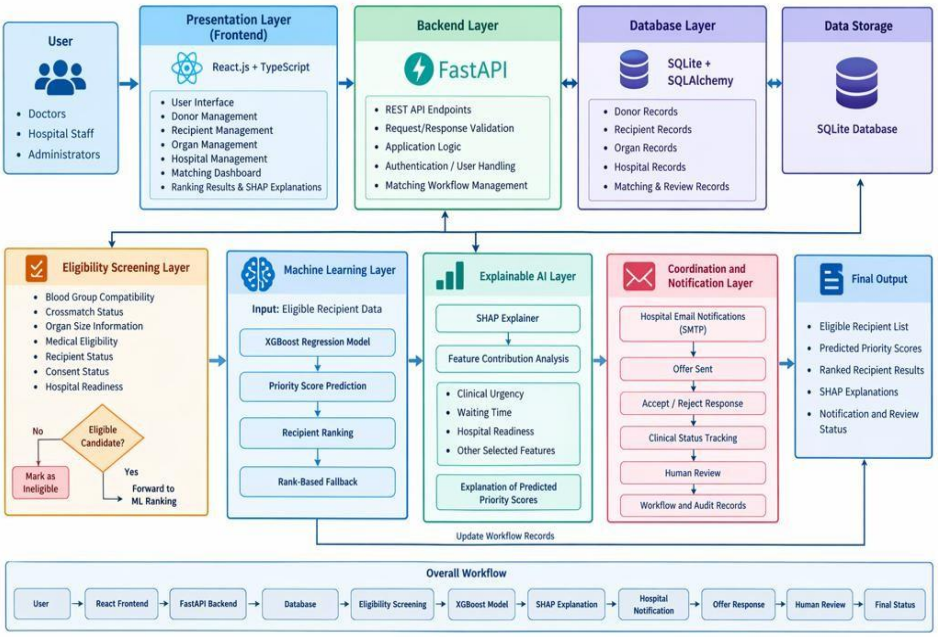


Fig. 4.1: Multi-Layer Architecture of the Proposed Organ Transplant Coordination System

The proposed system follows a multi-layer architecture, as shown in Fig. 4.1. Donor and recipient information is entered through the frontend and sent to the FastAPI backend. The backend performs eligibility checks, and eligible candidates are forwarded to the machine learning layer for priority-score prediction. SHAP provides explanations for the model predictions. Structured records are maintained in the database, while email notifications support hospital communication. Finally, the ranked results are presented for human review and further workflow processing.

4.3 System Components / Modules

The proposed system is divided into different modules to simplify development, testing, and maintenance. Each module performs a specific function in the organ-matching process.

4.3.1 Donor, Recipient, Organ, and Hospital Management

This module manages information related to donors, organs, recipients, and hospitals. It allows records to be created, retrieved, and updated for the matching process.

Donor records contain donor-related information, while organ records maintain details such as organ type and availability. Recipient records include selected matching features such as blood group, urgency category, and waiting time.

The module stores organized information in a relational database, allowing other system components to retrieve the required records during the matching process.

4.3.2 Rule-Based Eligibility Filtering

The rule-based eligibility filtering module identifies eligible donor-recipient combinations using predefined clinical and administrative criteria. The implemented checks include blood group compatibility, crossmatch status, organ size information, medical eligibility, recipient status, and consent status. Candidates who do not satisfy the required conditions are excluded from the ranking stage. This separation ensures that machine learning-based prioritization is performed only after eligibility screening. However, the implemented rules represent a prototype screening mechanism and do not provide a complete clinical compatibility assessment.

4.3.3 XGBoost-Based Recipient Ranking

Candidates who pass the eligibility checks are forwarded to the recipient-ranking module. An XGBoost regression model generates priority scores using selected input features. The features include clinical urgency, waiting-time score, distance, and hospital

readiness score. Candidates are ranked in descending order based on their predicted scores. This ranking supports the comparison and review of eligible candidates. The model output is intended to assist decision-making and must not be treated as a final organ-allocation decision.

4.3.4 SHAP-Based Explainability

The explainability module generates explanations for the predictions made by the XGBoost model using SHAP (SHapley Additive exPlanations). For individual predictions, SHAP values represent the contributions of input features to the predicted score. These explanations help users understand how clinical urgency, waiting time, distance, and hospital readiness influence the generated priority scores. SHAP improves the interpretability of the ranking results. However, explainability alone does not establish the clinical correctness or fairness of a prediction.

4.3.5 Hospital Notification Module

The hospital notification module supports communication through automated emails using the SMTP protocol. When the workflow reaches the notification stage, the system collects the required offer-related information and sends an email to the configured hospital email address. This module demonstrates how recipient-ranking results can be connected to a structured hospital communication process.

4.3.6 Human Review and Approval

The human review module allows authorized users within the prototype workflow to examine ranked candidates and their associated information. Reviewers can check eligibility details,

predicted priority scores, and SHAP explanations before updating the workflow status. The system also supports rank-based fallback processing when a higher-ranked candidate cannot proceed.

Actual organ allocation must follow the applicable official procedures and remain under the control of authorized medical professionals.

4.4 Eligibility Criteria and Ranking Features

The proposed system uses two categories of information: eligibility criteria and ranking features. Eligibility criteria determine whether a candidate can proceed to the ranking stage, while ranking features are provided to the XGBoost model to generate priority scores.

Table 4.3 presents the selected eligibility criteria and ranking features along with their purposes.

Table 4.3: Eligibility Criteria and Ranking Features

Feature	Purpose
Blood Group Compatibility	Checks initial blood group compatibility between the donor and recipient.
Crossmatch Status	Represents the available crossmatch result for compatibility screening.
Organ Size	Provides organ-size information for the implemented organ-size assessment.
Medical Eligibility	Indicates whether the candidate satisfies the defined medical eligibility conditions.
Recipient Status	Determines whether the recipient is available for consideration.
Consent Status	Indicates the consent information available for the matching process.
Clinical Urgency	Represents the urgency category or

Distance	Represents the distance-related feature used by the ranking model.
Hospital Readiness	Indicates the readiness score of the receiving hospital.

The eligibility criteria and ranking features are presented in Table 4.3. Eligibility criteria are applied before model prediction, whereas the selected ranking features are provided to XGBoost to generate priority scores. The meaning of each feature depends on the data and implementation used in the application.

4.5 Proposed Workflow

The proposed workflow begins with entering donor, organ, and recipient information into the web application. The backend retrieves the relevant records and applies the predefined eligibility rules. Candidates who satisfy the implemented eligibility criteria are forwarded to the XGBoost model. The model predicts priority scores based on the selected ranking features. SHAP analysis is then performed to explain how individual features influence the predicted scores. The eligible candidates are sorted based on their predicted scores and presented for human review. The notification module supports communication with hospitals, while the offer-response workflow tracks acceptance and rejection. Rank-based fallback processing may be used when a higher-ranked candidate cannot proceed, subject to appropriate review and authorization.

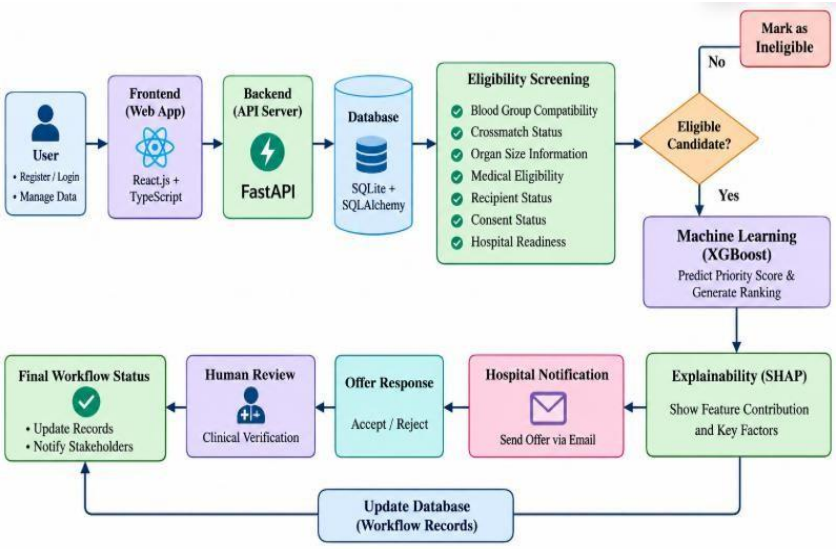


Fig. 4.2: Proposed Workflow of the Organ Transplant Coordination System

The proposed workflow is illustrated in Fig. 4.2. The process begins with donor and recipient information collection, followed by compatibility and eligibility checks. Eligible candidates are evaluated using XGBoost for priority prediction, while SHAP provides explanations for the predictions. The results are ranked for human review, followed by hospital notifications and offer-response processing. The workflow also supports status tracking and rank-based fallback handling when required.

4.6 Algorithms and Mathematical Formulations

The proposed system consists of the following main stages: rule-based eligibility filtering, feature preparation, XGBoost prediction, SHAP explanation, recipient ranking, and human review. The mathematical formulations used to describe these stages are presented below.

4.6.1 Rule-Based Eligibility Verification

The eligibility stage determines whether a candidate satisfies the implemented conditions required to proceed to the next stage.

Let the eligibility of candidate (i) be represented by (E_i).

$$E_i = \begin{cases} 1, & \text{if candidate } i \text{ satisfies all eligibility rules} \\ 0, & \text{otherwise} \end{cases} \rightarrow (1)$$

Where:

- (E_i) represents the eligibility status of candidate (i).
- (E_i = 1) indicates that the candidate has passed the implemented checks.
- (E_i = 0) indicates that the candidate has failed at least one required check.

Only candidates with (E_i = 1) proceed to the ranking stage.

4.6.2 Feature Preparation

Eligible candidates are represented using the selected priority-ranking features. The feature vector for candidate (i) is defined as:

$$X_i = [x_{i1}, x_{i2}, x_{i3}, \dots, x_{in}] \rightarrow (2)$$

Where:

- (U_i) represents the clinical urgency feature.
- (W_i) represents the waiting-time score.
- (D_i) represents the distance feature.
- (H_i) represents the hospital-readiness score.
- (X_i) represents the complete feature vector of candidate (i).

The features are prepared in the same format as required by the trained XGBoost model.

4.6.3 XGBoost Priority-Score Prediction

The XGBoost model assigns a priority score to each eligible candidate.

$$\hat{y}_i = \sum_{k=1}^{K} f_k(X_i), \quad f_k \in \mathcal{F} \rightarrow (3)$$

Where:

- (\hat{y}_i) represents the predicted priority score.
- (X_i) represents the feature vector of candidate (i).
- (f_k) represents an individual decision tree in the trained XGBoost model.
- (K) represents the number of trees.
- (F) represents the set of decision trees used by the model.

Candidates are ranked based on their predicted scores. A higher predicted score receives a higher rank in the prototype's configured ranking procedure. This ranking does not represent an official clinical priority-allocation policy.

4.6.4 Recipient Ranking

The predicted priority scores are used to arrange eligible candidates in descending order.

$$R = \text{Sort}(\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_n) \rightarrow (4)$$

Where:

- (R) represents the sorted list of eligible candidates.
- (\hat{y}_i) represents the predicted priority score of candidate (i).
- (n) represents the number of eligible candidates.

4.6.5 SHAP-Based Feature Contribution

SHAP is used to explain the contribution of each feature to the model prediction.

$$\phi_j = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!(|N|-|S|-1)!}{|N|!} [f(S \cup \{j\}) - f(S)] \rightarrow (5)$$

Where:

- (f(x)) represents the model prediction function.
- (\phi_j) represents the SHAP value of feature (j).
- (N) represents the set of all input features.
- (S) represents a subset of input features.

A positive SHAP value indicates that a feature contributes to increasing the prediction relative to the baseline, while a negative SHAP value indicates that it contributes to decreasing the prediction.

4.6.6 Model Evaluation Metrics

Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination ((R^2)) are used to evaluate the XGBoost model.

Mean Absolute Error (MAE):

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \rightarrow (6)$$

Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \rightarrow (7)$$

Coefficient of Determination ((R^2)):

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \rightarrow (8)$$

Where:

- (y_i) represents the actual target value.
- (\hat{y}_i) represents the predicted target value.

- (\bar{y}) represents the mean of the actual target values.
- (n) represents the number of evaluated samples.

Lower MAE and RMSE values indicate smaller prediction errors. An (R^2) value closer to 1 indicates that the model explains a greater proportion of the variation in the target values. These metrics evaluate prediction performance and do not determine clinical suitability.

4.6.7 Human-in-the-Loop Decision Review

In the final stage, the ranked list of eligible recipients, predicted priority scores, and explanations are presented for human review.

Let (D_i) represent the workflow review status of candidate (i).

$$D_i \in \{\text{Accept,Reject,Review}\} \rightarrow (9)$$

Where:

- (D_i) represents the review status of candidate (i).
- Accept indicates that the workflow offer or step has been accepted.
- Reject indicates that the workflow offer or step has been rejected.
- Review indicates that further examination is required.

4.7 Database Design

The proposed system uses an SQLite relational database and SQLAlchemy for database interaction. Structured information related to donors, recipients, organs, hospitals, matching activities, offers, and workflow records is stored in the database. The database is organized into separate entities to support structured storage and retrieval. Organ records contain information about organ types and availability, while donor records maintain donor-related details.

Recipient records store information about potential recipients. Matching records maintain details of the matching process and its results. Hospital response information and candidate status updates are maintained through offer and workflow records. The relational database structure supports organized data management and allows the backend application to retrieve the information required by different modules. The main database components and their purposes are presented in Table 4.4.

Table 4.4: Database Design

Entity / Table	**Main Information**	**Purpose**
Donor	Donor information	Maintains donor records.
Organ	Organ type and status	Stores organ-related information and availability status.
Recipient	Recipient details	Maintains recipient records.
Hospital	Hospital information	Stores hospital-related details.
Matching	Matching information and results	Maintains matching activities and their results.

Table 4.4 presents the logical database entities and their purposes. The database supports structured record management and retrieval for matching, ranking, notifications, and review activities.

The entity names, columns, and relationships in the final report should correspond to those implemented in the project source code.

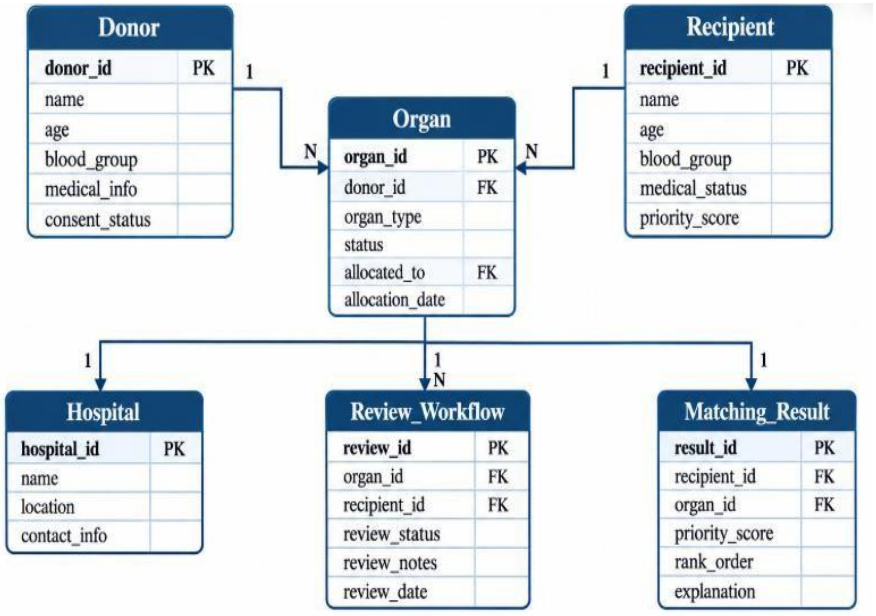


Fig. 4.3: Database Design of the Proposed Organ Transplant Coordination System

The database design of the proposed Organ Transplant Coordination System is illustrated in Fig. 4.3. The relationships between donor, organ, recipient, hospital, matching, offer, and review records support organized information storage and retrieval during the application's matching and workflow activities.

4.8 Coding / Program Implementation

The proposed organ matching and recipient ranking system is implemented as a full-stack web application. The frontend is developed using React.js and TypeScript, while Python and FastAPI are used for backend development. SQLAlchemy manages database operations, and SQLite is used for data storage. The XGBoost model predicts recipient priority scores, while SHAP explains the model predictions. Hospital email notifications are supported through SMTP.

The project is organized into frontend, backend, machine learning, database, and notification components. The major modules and their responsibilities are summarized in Table 4.5.

Table 4.5 File and Module Structure of the Organ Matching System

Path / Module	Description
Frontend	User interface
Backend	Application logic
Database	Data management
Matching & ML	Eligibility and recipient ranking
Explainability	Prediction explanations
Email Service	Hospital notifications
Configuration	Dependencies and settings

A summary of the major project components and their responsibilities is presented in Table 4.5. The frontend communicates with the backend through REST APIs. The backend coordinates database operations, eligibility processing, model prediction, explanation generation, and notifications. The main implementation stages include donor and recipient record management, rule-based eligibility filtering, feature preparation, XGBoost prediction, SHAP explanation, hospital communication,

and human review. The source-code listings included in the final report should be taken directly from the latest version of the GitHub repository. This ensures that function names, API routes, feature names, and implementation details match the submitted project.

The project source code is available in the following GitHub repository:

GitHub Repository: https://github.com/Parameshwari19/AI-Organ-Matching-System

LISTING 4.1: Rule-Based Eligibility Checking Function

```
from sqlalchemy import Column, Integer, String, Float, Boolean
from .database import Base
class Donor(Base):
    __tablename__ = "donors"
    donor_id = Column(String, primary_key=True, index=True)
    donor_type = Column(String)
    age = Column(Integer)
    sex = Column(String)
    height_cm = Column(Float)
    weight_kg = Column(Float)
    blood_group = Column(String)
    cause_of_death_category = Column(String)
    medical_history_category = Column(String)
    consent_status = Column(String)
    verification_status = Column(String)
    hospital_id = Column(String)
    donor_status = Column(String)
class Recipient(Base):
    __tablename__ = "recipients"
```

```
    age = Column(Integer)
    sex = Column(String)
    blood_group = Column(String)
    registration_date = Column(String)
    waiting_days = Column(Integer)
    urgency_category = Column(String)
    medical_risk_level = Column(String)
    medical_history_category = Column(String)
    active_status = Column(String)
    consent_status = Column(String)
class Organ(Base):
    __tablename__= "organs"
    organ_id = Column(String, primary_key=True, index=True)
    donor_id = Column(String)
    organ_type = Column(String)
    retrieval_datetime = Column(String)
    preservation_method = Column(String)
    organ_condition_category = Column(String)
    status = Column(String)
    current_hospital_id = Column(String)
class Hospital(Base):
    __tablename__= "hospitals"
    hospital_id = Column(String, primary_key=True, index=True)
    hospital_name = Column(String)
    city = Column(String)
    state = Column(String)
    latitude = Column(Float)
```

```
    longitude = Column(Float)
    hospital_type = Column(String)
    authorization_status = Column(String)
    transplant_center_status = Column(String)
    icu_available = Column(Boolean)
    operating_theatre_available = Column(Boolean)
    transplant_team_available = Column(Boolean)
    readiness_status = Column(String)
```

LISTING 4.2: Recipient Feature Preparation Function

```
def filter_eligible_candidates(match_runs):
    eligible_runs = []
    for run in match_runs:
        eligibility_conditions = [
            run.blood_group_compatible,
            run.crossmatch_compatible,
            run.size_compatible,
            run.recipient_active,
            run.recipient_consent_verified,
            run.hospital_authorized,
            run.medical_eligible
        ]
        if all(eligibility_conditions):
            eligible_runs.append(run)
    return eligible_runs
```

LISTING 4.3: XGBoost Model Loading and Priority Prediction

```
import os
import joblib
```

```
import shap
MODEL_PATH = os.path.join(
    os.path.dirname(os.path.dirname(__file__)),
    "organ_priority_xgboost.pkl"
)
model = joblib.load(MODEL_PATH)
explainer = shap.TreeExplainer(model)
def predict_recipient_priority(eligible_runs):
    feature_rows = [
        [
            run.urgency_score,
            run.waiting_time_score,
            run.distance_km,
            run.hospital_readiness_score
        ]
        for run in eligible_runs
    ]
    predictions = model.predict(feature_rows)
    candidates = []
    for run, prediction in zip(eligible_runs, predictions):
        candidates.append({
            "recipient_id": run.recipient_id,
            "organ_id": run.organ_id,
            "predicted_priority_score":  float(prediction)
        })
    candidates.sort(
        key=lambda item: (
```

```
            -item["predicted_priority_score"],
            item["recipient_id"]
        )
    )
    for index, candidate in enumerate(candidates, start=1):
        candidate["rank"] = index
    return candidates
```

LISTING 4.4: SHAP-Based Prediction Explanation

```
def explain_recipient_priority(eligible_runs):
    feature_rows = [
        [
            run.urgency_score,
            run.waiting_time_score,
            run.distance_km,
            run.hospital_readiness_score
        ]
        for run in eligible_runs
    ]
    shap_values = explainer.shap_values(feature_rows)
    explanations = []
    for index, run in enumerate(eligible_runs):
        candidate_shap = shap_values[index]
        explanations.append({
            "recipient_id": run.recipient_id,
            "shap_explanation": {
                "urgency_score": float(candidate_shap[0]),
                "waiting_time_score": float(candidate_shap[1]),
```

```
            "distance_km": float(candidate_shap[2]),
            "hospital_readiness_score": float(candidate_shap[3])
        }
    })
  return explanations
```

LISTING 4.5: SMTP-Based Hospital Email Notification

```
import os
import smtplib
from email.message import EmailMessage
from pathlib import Path
from dotenv import load_dotenv
BASE_DIR = Path(__file__).resolve().parent.parent
ENV_FILE = BASE_DIR / ".env"
load_dotenv(dotenv_path=ENV_FILE, override=True)
SMTP_SERVER = os.getenv("SMTP_SERVER",
"smtp.gmail.com")
SMTP_PORT = int(os.getenv("SMTP_PORT", "587"))
SMTP_USERNAME = os.getenv("SMTP_USERNAME")
SMTP_PASSWORD = os.getenv("SMTP_PASSWORD")
def send_email(recipient_email, subject, body):
  if not SMTP_USERNAME or not SMTP_PASSWORD:
    return {
      "success": False,
      "message": "SMTP credentials are missing"
    }
  message = EmailMessage()
  message["From"] = SMTP_USERNAME
```

```
message["To"] = recipient_email
message["Subject"] = subject
message.set_content(body)
try:
    server = smtplib.SMTP(
        SMTP_SERVER,
        SMTP_PORT,
        timeout=20
    )
    server.ehlo()
    server.starttls()
    server.ehlo()
    server.login(
        SMTP_USERNAME,
        SMTP_PASSWORD
    )
    server.send_message(message)
    server.quit()
    return {
        "success": True,
        "message": "Email sent successfully"
    }
except Exception as error:
    return {
        "success": False,
        "message": str(error)
    }
```

LISTING 4.6: FastAPI Recipient Ranking Endpoint

```
from datetime import datetime
from fastapi import HTTPException
def submit_clinical_review_decision(
    review_id,
    request,
    db
):
    decision = request.decision.strip().upper()
    if decision not in ("APPROVED", "NOT APPROVED"):
        raise HTTPException(
            status_code=400,
            detail="Decision must be APPROVED or NOT
APPROVED"
        )
    clinical_review = (
        db.query(models.ClinicalReview)
        .filter(
            models.ClinicalReview.review_id == review_id
        )
        .first()
    )
    if not clinical_review:
        raise HTTPException(
            status_code=404,
            detail="Clinical review not found"
        )
```

```
if clinical_review.status == "Completed":
    raise HTTPException(
        status_code=400,
        detail="Clinical review already completed"
    )
reviewer = request.reviewed_by.strip()
if not reviewer:
    raise HTTPException(
        status_code=400,
        detail="Reviewer name cannot be empty"
    )
clinical_review.reviewed_by = reviewer
clinical_review.review_date = datetime.now().isoformat()
clinical_review.decision = decision
clinical_review.remarks = request.remarks
clinical_review.status = "Completed"
db.commit()
db.refresh(clinical_review)
if decision == "APPROVED":
    allocation = finalize_approved_allocation(
        db,
        clinical_review
    )
    return {
        "success": True,
        "message": "Clinical review approved",
        "allocation": allocation
```

```
}
    next_result = notify_next_rank(
        db,
        clinical_review.organ_id,
        clinical_review.rank
    )
    return {
        "success": True,
        "message": "Clinical review was not approved",
        "next_notification": next_result
    }
```

The actual code for each listing must be copied from the corresponding implementation files in the repository. Placeholder descriptions should be replaced with the verified source code before submission. The implementation integrates compatibility checking, rule-based eligibility filtering, machine learning-based recipient ranking, explainable predictions, database management, and hospital communication. The XGBoost model generates priority scores, while SHAP provides feature-level explanations. The system also supports offer-response management, rank-based fallback, and human review. The proposed application is an academic prototype evaluated using synthetic data. It is not intended for real clinical organ allocation.

CHAPTER 3
RESULTS AND DISCUSSION

5.1 Introduction

This chapter presents the experimental setup, evaluation metrics, results, and discussion of the proposed Explainable AI-Powered Organ Transplant Coordination System. A synthetic dataset is used to evaluate the XGBoost-based recipient-ranking model. The chapter discusses the distribution of eligible and ineligible candidates, prediction performance, and feature contributions obtained through SHAP analysis.

5.2 Experimental Setup

The proposed system was developed using frontend and backend technologies. The frontend was built using React.js, TypeScript, and Vite, while the backend was developed using FastAPI and Python. SQLAlchemy was used for database management with SQLite. The machine learning component uses the XGBoost algorithm for recipient ranking, while SHAP is used to explain the model predictions. The experiment was conducted using a synthetic dataset containing 356 match runs. Rule-based eligibility screening identified 56 candidates as eligible and 300 candidates as ineligible. The eligible candidates were then processed further using the XGBoost ranking model.

Table 5.1: Dataset Overview

Description	Value
Total Match Runs	356
Eligible Candidates	56
Ineligible Candidates	300
Dataset Type	Synthetic
Ranking Algorithm	XGBoost
Explainability Technique	SHAP

The experimental setup demonstrates the integration of rule-based eligibility screening, machine learning-based ranking, and explainability within the proposed academic prototype.

5.3 Evaluation Metrics

Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination ((R^2)) were used to evaluate the performance of the XGBoost model. These metrics measure prediction errors and indicate how well the model explains variations in the target values.

Table 5.2: Performance Evaluation Metrics

Evaluation Metric	Obtained Value	Description
Mean Absolute Error (MAE)	0.0202	Measures the average absolute difference between actual and predicted values.
Root Mean	0.0310	Measures prediction error

Coefficient of Determination ((R^2))	0.9231	Indicates how much variation in the target values is explained by the model.

The evaluation results indicate that the model achieved relatively low prediction errors and an (R^2) value of 0.9231 on the synthetic experimental dataset. These values represent the performance of the experimental model and do not establish clinical validity.

5.4 Experimental Results

The experimental results demonstrate the performance of rule-based eligibility screening and XGBoost-based recipient ranking. Out of 356 match runs, 300 candidates were classified as ineligible based on the configured screening criteria, while 56 candidates were identified as eligible for further ranking.

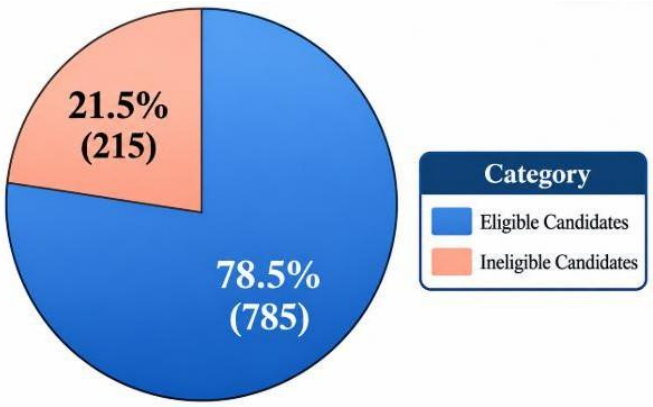


Fig. 5.1: Distribution of Eligible and Ineligible Candidates

Fig 5.1 illustrates the distribution of eligible and ineligible candidates identified during the experimental evaluation. The results show that 56 candidates passed the implemented eligibility criteria and were forwarded for further ranking, while 300 candidates did not satisfy the required conditions.

The XGBoost model achieved an MAE of 0.0202, an RMSE of 0.0310, and an (R^2) score of 0.9231.

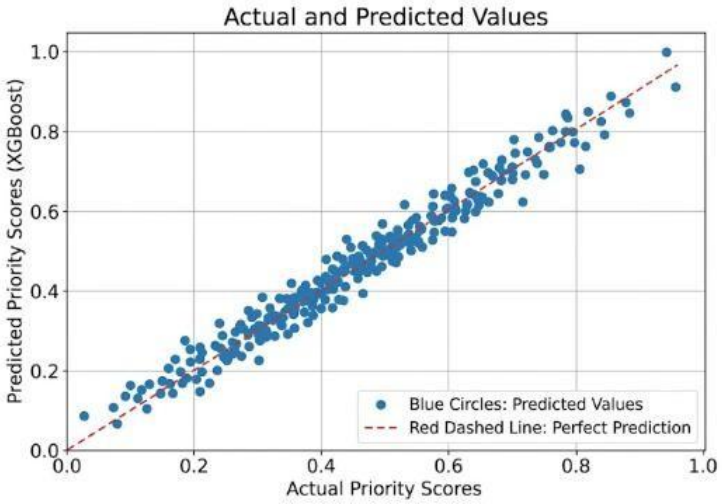


Fig. 5.2: Comparison of Actual and Predicted Values

Fig 5.2 presents the relationship between the actual target values and the values predicted by the XGBoost model. The comparison helps evaluate how closely the predicted values correspond to the actual values. The plot should be generated using the actual model evaluation data rather than estimated or manually created values.

5.5 Comparison with Existing Techniques

The proposed system integrates rule-based eligibility screening, machine learning-based recipient ranking, and explainable AI to support the organ-matching workflow. Traditional rule-based methods depend on predefined conditions, whereas machine learning models identify patterns in input data and generate

predictions. Explainability techniques such as SHAP provide additional information about the contribution of individual features to model predictions. Table 5.3 presents a comparison of the different techniques.

Table 5.3: Comparison with Existing Techniques

Technique	Main Function	Limitation/ Consideration
Traditional Machine Learning	Identifies patterns in data and generates predictions.	Predictions may be difficult to interpret without explanation techniques.
XGBoost	Predicts recipient priority scores using gradient boosting.	Requires appropriate training data and performance evaluation.
SHAP	Explains the contribution of input features to model predictions.	Explains model behaviour but does not establish clinical importance.
Proposed System	Combines eligibility screening, XGBoost ranking,SHAP explanations, notifications,and workflow tracking.	Currently evaluated using synthetic data and requires further validation.

The comparison highlights the integrated approach of the proposed academic prototype. It combines eligibility screening, recipient ranking, prediction explanations, and workflow support within a single application. However, it is not intended to replace official organ-allocation procedures or medical decisions made by authorized healthcare professionals.

5.6 Discussion of Results

The experimental evaluation demonstrates that the proposed system can perform rule-based eligibility screening and machine learning-based recipient ranking. A total of 356 match runs were evaluated, of which 56 were identified as eligible and 300 as ineligible. This indicates that the screening module separates candidates according to the implemented eligibility conditions.

The XGBoost model achieved an MAE of 0.0202, an RMSE of 0.0310, and an (R^2) score of 0.9231. These results indicate relatively low prediction errors on the synthetic dataset. However, the results are limited to the experimental environment and should not be directly applied to real-world organ transplantation.

SHAP analysis was used to understand the contribution of input features to the model predictions. The experimental feature-contribution analysis identified clinical urgency and hospital readiness as the two major contributors.

Table 5.4: SHAP Feature Contributions

Feature	Contribution
Clinical Urgency	48.15%
Hospital Readiness	25.25%
Other Features	26.60%
Total	100%

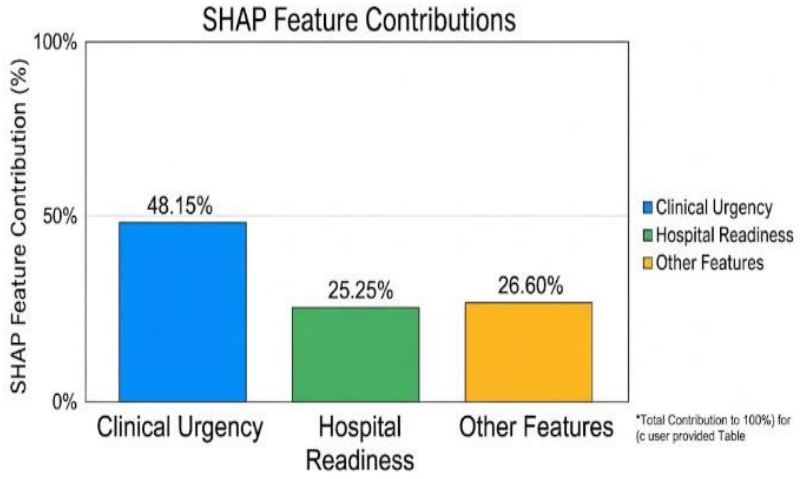


Fig. 5.3: SHAP Feature Contributions

Fig 5.3 illustrates the contribution of input features to the model predictions. Clinical urgency contributed 48.15%, while hospital readiness contributed 25.25%. The remaining features collectively contributed 26.60%. These percentages represent the feature contributions observed in the experimental dataset. They should not be interpreted as universal clinical weights or official organ-allocation priorities. Overall, the results demonstrate the technical operation of the proposed academic prototype. Any real-world application would require further evaluation using appropriate real-world data,approved institutional procedures.

5.7 Chapter Summary

This chapter presented the experimental setup, evaluation metrics, results, and discussion of the proposed Explainable AI-Powered Organ Transplant Coordination System. A synthetic dataset containing 356 match runs was used for evaluation, with 56 candidates identified as eligible and 300 as ineligible. The XGBoost model achieved an MAE of 0.0202, an RMSE of 0.0310, and an (R^2) score of 0.9231. SHAP analysis identified clinical urgency (48.15%) and hospital readiness (25.25%) as the two major feature contributors. The experimental results demonstrate the integration of eligibility screening, recipient ranking, explainability, and workflow support within the proposed academic prototype. However, these results do not establish clinical validity or suitability for real-world organ allocation.

CHAPTER 4

CONCLUSION AND FUTURE SCOPE

6.1 Summary of the Project

The project, titled “Explainable AI-Powered Organ Transplant Coordination”, is an academic prototype for the organ transplantation coordination process involving machine learning and Explainable Artificial Intelligence (XAI). It includes donor and recipient management, organ and hospital information management, rule based eligibility screening, XGBoost based recipient ranking, SHAP based explanations, hospital notifications and tracking of work done. The rule-based screening component is used to screen candidates for eligibility with the configured rules and the XGBoost model is used to generate priority scores to rank eligible recipient candidates. SHAP explanations are very useful to make the ranking process more transparent by revealing the contribution of the input features to the predictions made by the model. The notification and status-tracking part help with the communication and monitoring of the matching process. The system was tested with a synthetic set of data having a size of 356 match runs, 56 of which were identified as eligible candidates. The XGBoost model test results were: Mean Absolute Error (MAE) = 0.0202, Root Mean Squared Error (RMSE) = 0.0310, and R2 (coefficient of determination) = 0.9231. These results indicate the performance of the model in the data set.

6.2 Conclusion

The project illustrates the potential for the machine learning and explainable AI in an academic organ transplant coordination

prototype. The proposed system's benefits include its structured approach to organizing and monitoring the matching process, which is achieved through the rule-based eligibility screening, XGBoost-based recipient ranking, SHAP explainability, hospital notification support, and workflow tracking. The use of explainable AI helps users understand the factors contributing to the model's ranking predictions. The test data shows that the model works well with the synthetic data that was used for the evaluation. The system has not been clinically validated with real-world transplant data, however, and cannot determine clinical suitability or effectiveness of the allocation. As a whole the proposed system is designed merely as academic system to support and demonstrate the decision. It is not a real organ allocation system and does not replace the role of authorised medical specialists and does not replace the officially approved organ allocation procedures.

6.3 Future Scope

In future versions, the proposed system can be enhanced to better support the security, clinical review, organ-level tracking and hospital-to-hospital coordination. Authentication and authorization scheme can be designed in a hospital based system, where there are three different roles: Donor Hospital, Recipient Hospital, and Administrator. Several authorized accounts may be linked to each hospital, with JWT-based authentication, secure password storage, and Role Based Access Control (RBAC) to back this up. A deceased donor organ assessment module could also be created to show individual organs that will need additional assessment based on the available information from the donor and organ-specific

criteria. The assessment of each organ should be carried out separately and further coordination can be allowed only after proper consent and approval by authorized medical professionals.

One of the improvements is the addition of organ-specific tracking of the time in preservation. Different clinically approved preservation deadlines can be used for each approved organ. When the deadline of an organ is reached, its status can be automatically changed to expired, and it will be excluded from further matching and notification. You can also store expiry information in the audit history. Coordinated activities between donor hospitals, recipient hospitals and assigned medical personnel can be achieved using geographical information capabilities. The system can incorporate mapping and routing capabilities to calculate road distances and travel times, which can help with transportation planning and logistics. These estimates will only support logistical assistance and will not indicate the priority for allocation. Additional development may also involve the use of suitably controlled and clinically validated data sets to assess, and the implementation of enhanced security controls, additional audit facilities and expert assessment of model reliability, fairness, and generalization. Interconnection with approved healthcare information systems may be considered, as permitted by law, regulations, consent and institutional approvals. The proposed enhancements are designed to enhance the security, transparency, and coordination aspects of the proposed framework, all while maintaining the critical role of authorized medical professionals in every clinical decision.

SDG GOALS

SDG 3 – Good Health and Well-Being

Organ transplantation can save the lives of patients suffering from end-stage organ failure. By supporting recipient eligibility screening, priority ranking, and transparent decision-making, the proposed system explores how AI can assist organ-matching workflows. Human clinical review remains essential, as the system is an academic prototype and does not make real-world transplantation decisions.

SDG 9 – Industry, Innovation and Infrastructure

The project applies artificial intelligence, machine learning, and explainable AI to develop a structured organ-matching decision-support framework. By integrating rule-based filtering, XGBoost ranking, SHAP explanations, and automated hospital notifications, the system demonstrates an innovative approach to improving workflow organization and transparency. It provides a foundation for further research into responsible AI applications in healthcare infrastructure.

REFERENCES

[1] J. Allgaier, L. Mulansky, R. L. Draelos, and R. Pryss, "How does the model make predictions? A systematic literature review on the explainability power of machine learning in healthcare," Artificial Intelligence in Medicine, vol. 143, Art. no. 102616, 2023. doi: 10.1016/j.artmed.2023.102616.

[2] H. Ali, A. Shroff, T. Fülöp, M. Z. Molnar, A. Sharif, B. Burke, S. Shroff, D. Briggs, and N. Krishnan, "Artificial intelligence assisted risk prediction in organ transplantation: a UK Live-Donor Kidney Transplant Outcome Prediction tool," Renal Failure, vol. 47, no. 1, Art. no. 2431147, 2025. doi: 10.1080/0886022X.2024.2431147.

[3] N. Alowidi, R. Ali, M. Sadaqah, and F. M. A. Naemi, "Advancing Kidney Transplantation: A Machine Learning Approach to Enhance Donor-Recipient Matching," Diagnostics, vol. 14, no. 19, Art. no. 2119, 2024. doi: 10.3390/diagnostics14192119.

[4] "Artificial intelligence–driven kidney organ allocation: systematic review of clinical outcome prediction, ethical frameworks, and decision-making algorithms," BMC Nephrology, vol. 26, Art. no. 639, 2025. doi: 10.1186/s12882-025-04576-4.

[5] N. Börner, M. B. Schoenberg, P. Pöschke, C. Heiliger, S. Jacob, D. Koch, B. Pöllmann, M. Drefs, D. Koliogiannis, C. Böhm, K. W. Karcz, J. Werner, and M. Guba, "A Novel Deep Learning Model as a Donor-Recipient Matching Tool to Predict Survival after Liver Transplantation," Journal of Clinical Medicine, vol.

11, no. 21, Art. no. 6422, 2022. doi: 10.3390/jcm11216422.

[6] E. M. Bunnik, "Ethics of allocation of donor organs," Current Opinion in Organ Transplantation, vol. 28, no. 3, pp. 192–196, 2023. doi: 10.1097/MOT.0000000000001058.

[7] R. Dale, M. Cheng, K. Casselman Pines, and M. E. Currie, "Inconsistent values and algorithmic fairness: a review of organ allocation priority systems in the United States," BMC Medical Ethics, vol. 25, Art. no. 115, 2024. doi: 10.1186/s12910-024-01116-x.

[8] R. Deshpande, "Smart match: revolutionizing organ allocation through artificial intelligence," Frontiers in Artificial Intelligence, vol. 7, Art. no. 1364149, 2024. doi: 10.3389/frai.2024.1364149.

[9] S. Ding, D. Zha, K. Zhang, L. Chen, X. Jiang, X. Hu, and N. Zou, "FairAlloc: Learning Fair Organ Allocation Policy for Liver Transplant," Journal of Healthcare Informatics Research, vol. 9, no. 4, pp. 703–719, 2025. doi: 10.1007/s41666-025-00206-8.

[10] "Ensemble-Based Predictive Modeling Machine Learning Framework for Live-kidney Donation Prediction," Procedia Computer Science, vol. 260, pp. 476–485, 2025. doi: 10.1016/j.procs.2025.03.225.

[11] S. Gnanasambandan and V. Balasubramanian, "A data-driven framework for fair and efficient organ transplantation using gradient boosting and adaptive genetic allocation," Journal of Artificial Organs, vol. 28, no. 4, pp. 527–537, 2025. doi: 10.1007/s10047-025-01512-z.

[12] A. Lebret, "Allocating organs through algorithms and equitable access to transplantation—a European human rights law approach," Journal of Law and the Biosciences, vol. 10, no. 1, Art. no. lsad004, 2023. doi: 10.1093/jlb/lsad004.

[13] C. Li, D. Lai, X. Jiang, and K. Zhang, "FERI: A Multitask-based Fairness Achieving Algorithm with Applications to Fair Organ Transplantation," AMIA Joint Summits on Translational Science Proceedings, pp. 593–602, 2024.

[14] "Machine Learning Algorithms in Controlled Donation After Circulatory Death Under Normothermic Regional Perfusion: A Graft Survival Prediction Model," Transplantation, vol. 109, no. 7, pp. e362–e370, 2025. doi: 10.1097/TP.0000000000005312.

[15] S. A. A. Naqvi et al., "Donor-Recipient Matching for Kidney Transplantation using Uncertainty Estimation in Generalized Propensity Score," Studies in Health Technology and Informatics, 2025. doi: 10.3233/SHTI250395.

[16] D. B. Olawade, S. Marinze, N. Qureshi, K. Weerasinghe, and J. Teke, "The impact of artificial intelligence and machine learning in organ retrieval and transplantation: A comprehensive review," Current Research in Translational Medicine, vol. 73, no. 2, Art. no. 103493, 2025. doi: 10.1016/j.retram.2025.103493.

[17] L. Pruinelli, K. Balakrishnan, S. Ma, Z. Li, A. Wall, J. C. Lai, J. D. Schold, T. Pruett, and G. Simon, "Transforming liver transplant allocation with artificial intelligence and machine learning: a systematic review," BMC Medical Informatics and Decision Making, vol. 25, Art. no. 98, 2025. doi: 10.1186/s12911-025-02890-3.

[18] J. Rueda et al., "'Just' accuracy? Procedural fairness demands explainability in AI-based medical resource allocations," AI & Society, vol. 39, pp. 1411–1422, 2024. doi: 10.1007/s00146-022-01614-9.

[19] A. Salaün, S. Knight, L. Wingfield, and T. Zhu, "Predicting graft and patient outcomes following kidney transplantation using interpretable machine learning models," Scientific Reports, vol. 14, Art. no. 17356, 2024. doi: 10.1038/s41598-024-66976-0.

[20] "TransplantPredictor: A deep learning framework for liver transplant matching and survival prediction," Results in Engineering, 2025. doi: 10.1016/j.rineng.2025.107351.

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